

First Thesis Final Draft

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1 Abstract

Life as we know it cannot survive in a hydrogen and helium dominated environment that is a primordial atmosphere. We investigate the mass loss rate of an exoplanet’s primordial atmosphere with the goal of determining what conditions are necessary for an exoplanet in the habitable zone to lose its first atmosphere in preparation for a potentially habitable second. To estimate atmospheric mass loss via hydrodynamic escape, we use the energy-limited approximation and vary stellar mass and rotation rate. Then, we identify the effects of stellar XUV luminosity evolution on the time needed for a $0.5M_{\oplus}$ sub-Earth to lose its first atmosphere. We consider stars ranging from $0.3 - 1.0M_{\odot}$ whose rotation rates are drawn from the rotational evolution distribution from Johnstone et al. (2021). We find that the combination of the smallest star with the fastest rotation results in the shortest time scale over which the atmosphere fully evaporates. We reason that the shorter orbital distance to the habitable zone in tandem with the higher XUV luminosity due to fast rotation causes faster photoevaporation of the atmosphere. We then expand this study to include more sub-Earth planets ranging in mass from $0.1 - 1.0M_{\oplus}$, each with a H/He gaseous envelope 0.5% of their core masses. We conclude that the smallest planet retains its atmosphere for less time than a more massive counterpart because of its lower gravity and lower initial atmospheric mass. We encourage future work to utilize hydrodynamics, planetary interior structure evolution, and planetary migration to more accurately account for changes in planetary radii and orbital separation.

2 Introduction

Primordial atmospheres are the first gaseous envelopes surrounding a planet, acquired during its formation. However, it is the secondary atmosphere that is more important for habitability as the first’s composition is not suitable for life as we know it. Over time, the primordial atmosphere is lost to space, primarily through hydrodynamic escape where intense heating (typically from stellar XUV radiation) causes outward, fluid-like flow, resulting in bulk atmospheric loss especially of light gases like hydrogen and helium (e.g. Watson et al. (1981); Erkaev et al. (2007); Owen and Alvarez (2015)). Of particular importance is the time taken for a planet to lose its primordial atmosphere because it is believed that a secondary atmosphere suitable for life must be given sufficient time to develop for life to form. In investigating said timescale of primordial atmospheres, we pose and answer the question:

How does hydrodynamic escape shape the evolution of primordial atmospheres on terrestrial exoplanets in the habitable zone?

With the main driver of hydrodynamic escape being XUV flux, we focus on three main goals while addressing our above research question:

1. Quantify how XUV-driven hydrodynamic escape removes primordial H/He from terrestrial planets in the habitable zone.
2. Map how escape rates vary with stellar mass, planetary mass, and XUV history.
3. Evaluate under what conditions terrestrial planets in the habitable zone retain or lose their primordial atmospheres and their implications for habitability.

2.1 Primordial Atmospheres

First, to accurately model how primordial atmospheres evolve, it is necessary to ascribe them an accurate composition; their compositions mirror that of the solar nebula at the time of the planet and solar system’s formation (e.g. Aller (1960); Prinn (1982)). This results in solar-like hydrogen- and helium- dominated atmospheres as they are acquired through the gravitational binding of nebular gas from the surrounding hydrogen- and helium-dominated protoplanetary disk (Lammer et al. (2018)). Therefore, in this report we ascribe H/He gaseous envelopes accounting for 0.5% of the mass of the planet core they encase. The transience of these H/He-rich envelopes is due to thermal escape processes described in §2.4. After their evaporation, secondary atmospheres take their place. Their formation and importance to habitability are discussed in §5.

2.2 Stellar Activity and Evolution

The evolution of primordial atmospheres depends on the evolution and activity of the planet’s host star. Low-mass stars like the Sun go through a pre-main-sequence phase (~ 10 -100 Myr) with high activity. Their rotation rates determine magnetic dynamo strength and thus XUV (X-ray and Extreme Ultraviolet) luminosity L_{XUV} , with rapid rotators staying active longer with more intense winds and radiation (e.g. Ribas et al. (2005); Güdel (2007)). XUV radiation is absorbed in a planet’s upper atmosphere and drives thermal escape, especially hydrodynamic escape of H/He as discussed in §2.4. The early evolution of a star’s XUV luminosity is characterized by a saturation phase whose duration varies based on the star’s mass and rotation rate. The ratio between L_{XUV} and the bolometric luminosity L_{bol} is approximately constant and at its highest during this phase with $L_{XUV}/L_{bol} \approx 10^{-3}$, shown in Figure 1. Once the saturation time is reached, the XUV output decreases as a power law over time as the star spins down as shown by Tu et al. (2015).

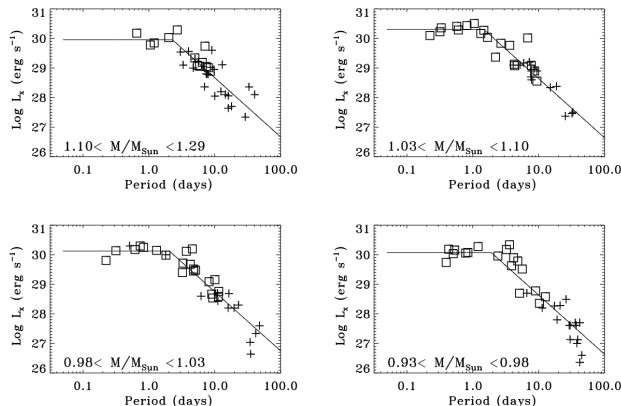


Figure 1: The X-ray to bolometric luminosity ratio vs. the rotation period for stars in varying mass ranges is shown. Squares indicate stars in clusters and crosses indicate field stars with measured periods. The broken power law that defines the transition from the saturated regime to the unsaturated regime is qualitatively identifiable and indicated by the solid line. This figure is adapted from Pizzolato et al. (2003).

This long-term evolution is closely tied to the star’s initial rotation rate, which depends on a combination of pre-main-sequence angular momentum evolution processes. For example, the Ghosh & Lamb (1978,1979) model is a framework describing the interaction between a magnetized, rotating star and its accretion disk. It was later adapted to T Tauri stars by Koenigl (1991) where it was first referred to as ”disk-locking”. Disk-locking is when young stars in the T Tauri phase are born with strong magnetic fields whose line thread the inner edge of the accretion disk. The inner disk then truncates at a few stellar radii where magnetic pressure balances gas pressure. Magnetic field lines transfer angular momentum between the star and the inner disk, preventing the star from spinning up as it contracts. If the disk persists longer, it results in a slow rotator, while it results in a fast rotator if it dissipates earlier. Furthermore, a higher accretion rate leads to faster stellar rotation as shown by Matt et al. (2010). The initial rotation rate of a star is indicative of its evolution as in Tu et al. (2015). Johnstone et al. (2021) presented the stellar rotation model that is used throughout

§3 and §4. The implementation of these stellar rotation models is described in further detail in §?? and are presented in Figure 2.

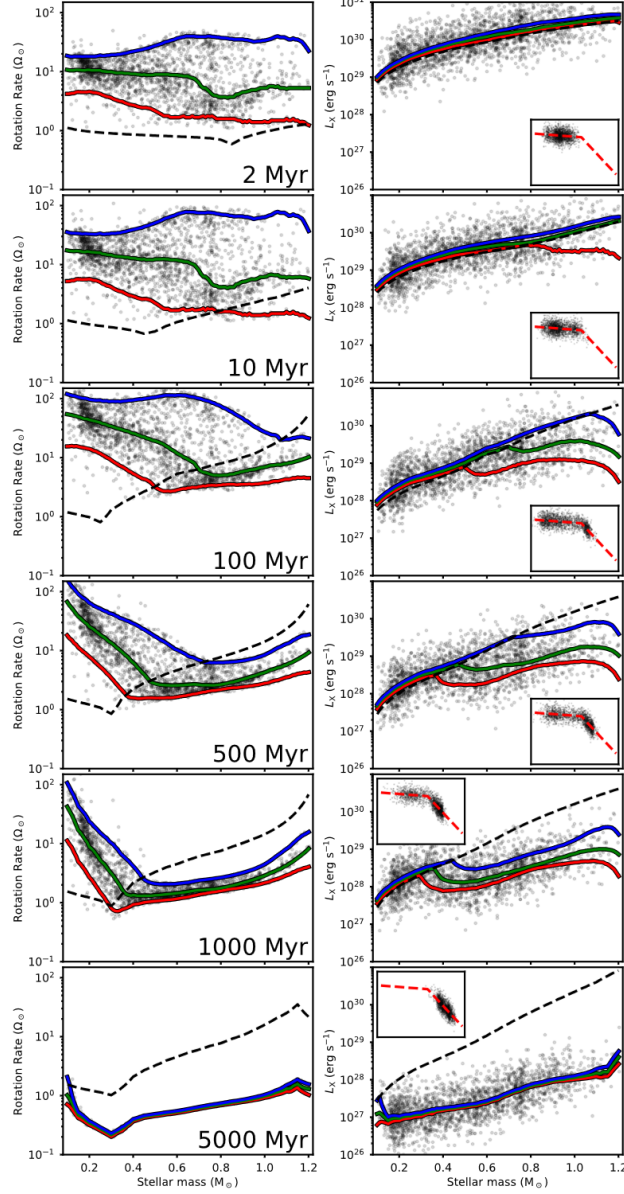


Figure 2: The left column shows that stars spin down over time and that smaller stars tend to have higher rotation rates than their more massive counterparts. The right column shows that stellar X-ray luminosity decreases over time. The red, green, and blue lines indicate the values for the 5th, 50th, and 95th percentile rotators drawn from the 150 Myr rotation distribution developed from several young clusters. These two columns are adapted from Johnstone et al. (2021).

2.3 Habitable Zone

As stellar luminosity evolves over time, it directly affects the location of the habitable zone (HZ) – the region around a star where conditions permit liquid water to exist on a planet’s surface (Kasting et al. (1993); Kopparapu et al. (2013)). This work uses the formulae of Kopparapu et al. (2013) in conjunction with the stellar models from Spada et al. (2013) to calculate the habitable zone boundaries over time. The location of the HZ depends on the host star’s spectral type and luminosity evolution as investigated by Ribas et al.

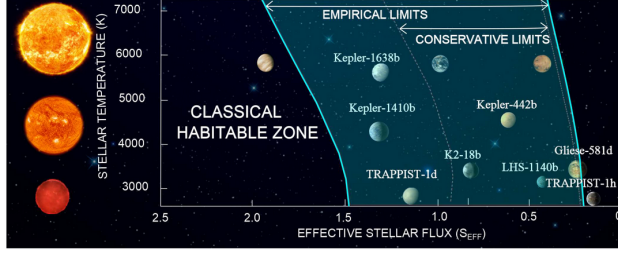


Figure 3: The classical habitable zone defined by the optimistic, empirical recent Venus and early Mars limits (solid blue curves; left to right) compared to the more conservative and theoretical moist and maximum greenhouse limits (dashed blue curves). Some confirmed exoplanets and solar system planets are depicted based on their orbital separations. Adapted from Ramirez (2018), the x-axis is defined as the effective stellar flux incident on the planet and the y-axis the stellar effective temperature.

(2005) and Johnstone et al. (2021). M dwarfs' HZs are closer in, exposing planets to stronger stellar winds and XUV fluxes (Johnstone et al. (2021)). The inner boundary of the empirically defined HZ in this work is the "recent Venus" limit based on the extrapolation that water has been absent from Venus's surface for at least the past 1 billion years (Solomon and Head (1991); Kopparapu et al. (2013)). The outer boundary of the HZ is the "early Mars" limit which is estimated using the deduction that surface water was present on Mars 3.8 billion years ago (Kasting et al. (1993); Kopparapu et al. (2013)). The other HZ limits referred to in this work are the theoretical moist- and maximum-greenhouse limits formulated by Kopparapu et al. (2013) where the moist-greenhouse limit is defined by the stellar flux at which stratospheric water vapor becomes abundant, leading to rapid water loss, and the maximum greenhouse limit is the orbital distance at which CO_2 no longer warms the planet. The empirically and theoretically defined habitable zones along with the orbital locations of discovered exoplanets are depicted in Figure 3.

2.4 Atmospheric Escape

Although the habitable zone defines where liquid water could exist, a planet's actual habitability depends on its capacity to retain an atmosphere against thermal and non-thermal escape processes. In this work, it is assumed that hydrodynamic escape, a thermal mechanism, is the driving force behind atmospheric mass loss. When high-energy photons are absorbed in the upper atmosphere, they heat the gaseous envelope, increasing thermal energy. The critical condition for hydrodynamic escape is $\frac{kT}{m} \gtrsim \frac{GM_p}{R_p}$ where T is the temperature of the upper atmosphere, m is the particle mass, G is the gravitational constant, and M_p and R_p are the mass and radius of the planet (e.g. Parker (1958); Watson et al. (1981)). If this energy is high enough, it drives an outward pressure gradient causing the atmosphere to expand and light species to escape the planet's gravity in bulk flow. Hydrodynamic escape can be estimated using the energy-limited approximation as discussed in §3.2 and is depicted in Figure 4.

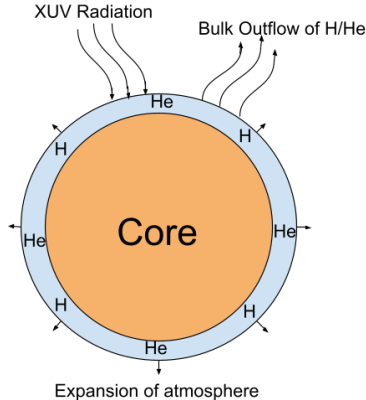


Figure 4: A diagram of a primordial H/He atmosphere (shown in blue) enveloping the planet’s core (shown in orange) and undergoing hydrodynamic escape driven by stellar XUV radiation. The small outward-pointing arrows indicate the expansion of the XUV-energized atmosphere that eventually leads to the bulk outflow of H/He indicated by the associated arrows and label. The XUV radiation incident on the planet is indicated by the associated arrows and label while the composition of the H/He atmosphere is indicated by the H and He present in the blue region.

Other thermal mechanisms that drive atmospheric escape include Jeans escape and boil-off. The former is a process in which the atmosphere is treated as a pool of individual particles rather than a fluid (Jeans (1916)). The individual particles gain enough thermal energy to exceed the planet’s escape velocity and are lost to space. Identified by Volkov et al. (2011), when the ratio of the gravitational to thermal energy at the lower boundary of the exosphere, known as the Jeans parameter λ_0 , is less than ~ 2 , hydrodynamic escape dominates. They also found that if $\lambda_0 > \sim 6$ Jeans escape dominates where $\lambda_0 = \frac{v_{esc}^2}{2v_{th}^2}$ and v_{esc} is the escape velocity while v_{th} is the most probable thermal speed of particles. In this work, thermal energy dominates, resulting in hydrodynamic escape being the dominant contributor to atmospheric mass loss. The sustained high XUV flux during the stellar saturation phase in combination with the low density, hydrogen-rich primordial atmospheres of the planets considered allows for thermal energy to be taken as dominant (e.g. Ribas et al. (2005), Tu et al. (2015), Lammer et al. (2014), Owen and Wu (2013)).

In addition to Jeans escape, boil-off escape also causes atmospheric mass loss. Occurring before stellar XUV-driven hydrodynamic escape takes over and immediately after protoplanetary disk dispersal, a planet’s atmosphere is thermally inflated and can spontaneously flow off due to excess internal and residual thermal energy. Typically lasting < 10 Myr after disk dispersal, boil-off escape is valid when the atmosphere is still cooling from formation and not yet gravitationally bound (e.g. Owen and Wu (2016), Ginzburg et al. (2018), Fossati et al. (2017)).

Beyond thermal escape, a variety of non-thermal mechanisms can also drive atmospheric loss. A primary example is impact erosion. During the first ~ 100 Myr, large impacts deliver kinetic energy that can eject atmospheric gas as shown in the case of Mars by Melosh and Vickery (1989). Next, photochemical escape involves UV photons dissociating molecules, producing energetic atoms that escape the planet’s gravitational pull. Photochemical escape is particularly relevant for secondary atmospheres rather than primaries because it acts on those gases supplied by outgassing (Lammer et al. (2008)). Furthermore, ion pickup escape which occurs from $\sim 1 - 10$ Myr onward when solar wind reaches the planet and atmospheric neutrals are ionized and picked up by the stellar wind’s electric field. The required interaction between stellar wind and the ionosphere is weak in early thick H/He atmospheres due to low ionization levels, resulting in escape rates much lower than those resulting from hydrodynamic escape (e.g. Moore and Horwitz (2007), Lammer et al. (2008)). Sputtering occurs from $\sim 10 - 100$ Myr onward when high-energy ions can reach the upper atmosphere and collide with atmospheric atoms, sending them into space. However, sputtering requires a thin, collisionless upper atmosphere while the atmospheres considered in this work are thick, hydrogen atmospheres. Lastly, ambipolar escape, or polar wind, occurs when ionized particles flow out along magnetic field lines from polar regions due to electric field as described by Schunk (1986) and Yau et al. (2007).

2.4.1 The Effect of Atmospheric Escape on the Habitable Zone

Simply being in the classically defined habitable zone is not sufficient for habitability; planetary atmospheres must be retained long enough for surface liquid water and life. The calculation of the atmospheric mass-loss rate as a result of hydrodynamic escape is presented in §3.2 to explore atmosphere retention. Photoevaporation may strip H/He envelopes entirely from young planets as shown throughout this work, potentially exposing rocky planetary cores and altering their classification and habitability potential (Lopez and Fortney (2012); Owen and Wu (2016)). Habitability of these planets is then dependent on whether they can retain or regenerate secondary atmospheres, a notion explored in §5.4. Moreover, the escape of key volatiles such as water vapor and CO₂ can cause a runaway greenhouse effect or make them too cold to sustain liquid water (Tian et al. (2005); Luger and Barnes (2015)). Therefore, atmospheric retention, not just orbital separation, is also a key boundary condition for defining habitability. Modern models of the habitable zone increasingly incorporate this consideration, acknowledging that habitability is ultimately a coupled function of stellar activity, planetary mass and composition, and atmospheric escape physics (e.g. Owen (2019), Meadows et al. (2018)). In §4, the time taken for planets’ primordial atmospheres to fully escape is investigated as a function of stellar and planetary parameters under the assumption that a planet must have a secondary atmosphere for liquid water to be present and therefore for life as we know it to be possible.

2.5 Scope and Objectives

Many previous studies that modeled atmospheric escape focused on close-in sub-Neptunes (Owen (2019); Luger and Barnes (2015)) or the rocky planets of our solar system (Lammer et al. (2015); Kislyakova et al. (2020); Gillmann et al. (2022)). Fewer have focused on terrestrial-mass planets specifically in the habitable zone over Gyr timescales. Key open questions include whether low-mass habitable zone planets can retain their primordial H/He atmospheres and what roles stellar mass and activity play in regulating long-term atmospheric mass loss.

This work models hydrodynamic escape of primordial atmospheres from terrestrial-mass planets located within the habitable zone, using MORS code framework (Johnstone et al. (2021)) explained in 3.1. Where this work differs from others is in its consideration of sub-Earth exoplanets and stellar evolution tracks. Planet cores in the mass range of $0.1 - 2.0 M_{\oplus}$ with initial gaseous envelopes of masses 0.5% of their core masses were considered. Only thermal escape, specifically hydrodynamic escape, driven by stellar XUV radiation is taken into account. Over a timescale of 1 Myr - 5 Gyr, the rotation rate of the host star evolves according to its initial mass and rotation rate, varying its XUV radiation over time. The stellar masses in this work range from $0.3 - 1.0 M_{\odot}$.

3 Methods

To quantify how XUV-driven hydrodynamic escape removes primordial atmospheres from habitable zone exoplanets, we use Model for Rotation of Stars (MORS), the Continuously Habitable Zone (CHZ), and the planetary mass-radius relations from Chen and Kipping (2016) to compute the atmospheric mass loss rate with the energy-limited approximation. MORS is a Python-based tool by Colin P. Johnstone that numerically solves the stellar rotation and XUV evolution framework introduced in Johnstone et al. (2021). In addition to XUV luminosity, the energy-limited approximation depends on the orbital separation between a star and its planet. This value is determined using CHZ, defined in this work as the region around a star that is habitable from the stellar ages of 100 Myr - 5 Gyr. Furthermore, the mass, radius, and radius where the planet receives XUV flux are also used to calculate a planet’s atmospheric mass-loss rate. These values are calculated with an adaptation of the planetary mass-radius relations from Chen and Kipping (2016). We then use the middle of the CHZ around each star as the orbital separation, the evolution of the XUV luminosity tracks from MORS, and the characteristics of the planets to calculate the atmospheric mass-loss rate from the energy-limited approximation.

3.1 Utilizing MORS

The basic procedure of using MORS in this paper involves the user defining the mass of the star and the initial rotation rate indicated by the percentile of the rotation evolution model distribution and then ages them from 1 million years to 5 billion years. The rotational model distribution is based on measured stellar rotation data from several clusters age ~ 150 million years evolved back to 1 million years. The EUV luminosity and X-ray luminosity are extracted at every millionth year to calculate the atmospheric mass loss rate. We verify consistent usage of MORS by comparing the X-ray flux in the HZ to that calculated by Johnstone et al. (2021), shown here in Figure 5. MORS solves the stellar rotation and XUV evolution model presented by Johnstone et al. (2021). The consistent output verifies that MORS was used by this study correctly, because for the X-ray flux in the habitable zone to be consistent across the plots, they used the same stellar XUV tracks, implying that the same XUV evolution model was solved by each.

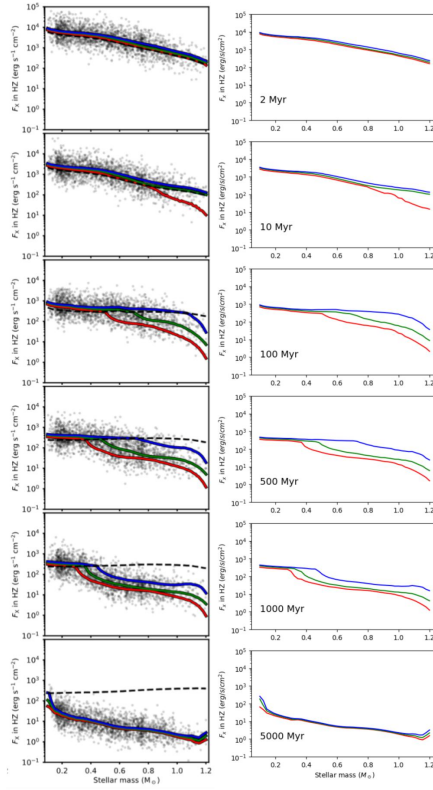


Figure 5: A display of consistency between Johnstone et al. (2021) and this work. The left column is adapted from Johnstone et al. (2021) and shows the evolution of X-ray flux in the habitable zone against stellar mass at different ages. The right column is a product of this work and shows the same evolution. In these cases, the habitable zone is defined as the distance halfway between the moist and maximum greenhouse limits. The red, green, and blue lines indicate the values for the 5th, 50th, and 95th percentile rotators (Slowest, Median, and Fastest as indicated in Table 1) from the 150 Myr rotation distribution composed of several clusters from Johnstone et al. (2021). In the left column, the black dashed line indicates the saturation threshold values, or the points at which further increases in rotation do not increase XUV emission, and the black circles show the model distribution. The ages in the right column apply to the ages of the stars across the rows in both columns.

Adjective	Rotation Distribution Percentile
Slowest	5th
Median	50th
Fastest	95th

Table 1: The quantitative values associated with the adjectives used throughout this report, shaded by the colors representing them in succeeding figures. The percentile of the rotational evolution model distribution from Johnstone et al. (2021) is specified in the right column with higher percentiles indicating faster initial stellar rotation.

Next, the results of Tu et al. (2015) were reproduced with modified input data to test the sensitivity of the model and to evaluate consistency across different stellar evolution tracks. Tu et al. (2015) made use of the stellar evolution models presented in Johnstone et al. (2015). This work uses the updated evolutionary track models presented in Johnstone et al. (2021). A comparison between the X-ray luminosities L_X and EUV luminosities L_{EUV} of a solar-mass star produced by Tu et al. (2015) and myself is presented in Figure 6.

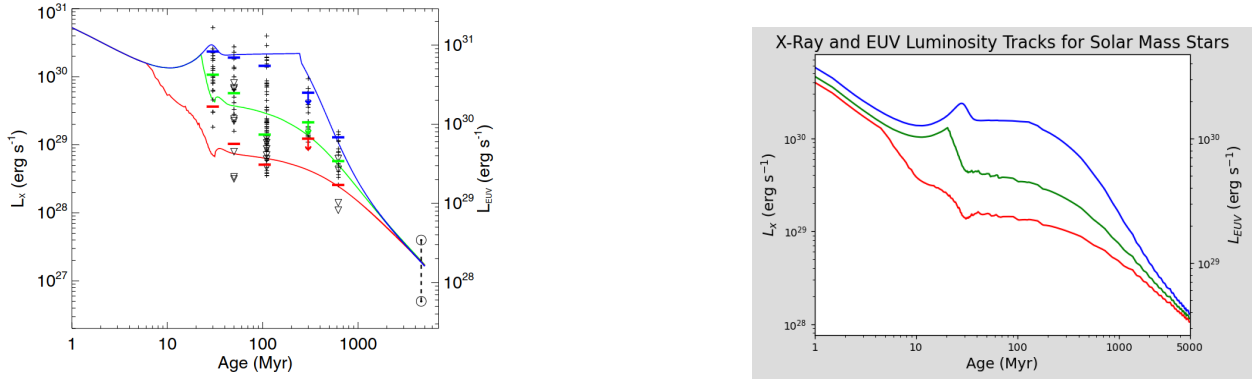


Figure 6: A comparison between the evolution of L_X and L_{XUV} for a solar-like star from the age of 1 Myr to 5 Gyr. The percentiles from the Johnstone et al. (2015) rotational distribution used in the left panel adapted from Tu et al. (2015) are the 10th, 50th, and 90th, while in the right panel they are consistent with what is used in the rest of the report: the 5th, 50th, and 95th percentiles from the rotational distribution from Johnstone et al. (2021).

3.2 Energy-Limited Approximation for Atmospheric Escape

In this work, the atmospheric mass loss rate is estimated with the energy-limited approximation. As explained by Krenn et al. (2021), this approach assumes that the atmosphere is hydrogen dominated and that hydrodynamic escape incited by XUV stellar radiation is the primary atmospheric escape mechanism. The mass-loss rate as a result of the energy-limited approximation $\dot{M}_E$ is given by:

$$\dot{M}_E = \epsilon \pi \frac{F_{XUV} R_{XUV}^2 R_{pl}}{GM_{pl}} \quad (1)$$

where ϵ is the fixed heating efficiency and assumed to be equal to 0.3¹, F_{XUV} is the incident stellar XUV flux on the planet absorbed at the radius from the planet's center where it is received R_{XUV} . R_{XUV} is in terms of the radius of the planet R_{pl} and characterizes how far the atmosphere extends above the surface of the planet. Each planet's gaseous envelope is assumed to extend past the surface by 10% of its core's

¹This value is on the upper end of theoretical upper bound for efficiency in hydrogen-dominated atmospheres (Salz et al. (2016); Odert, P. et al. (2020)) as the often taken range is 0.1 - 0.3 (e.g Owen and Wu (2013); Kubyshkina et al. (2018))

radius². Lastly, G is the gravitational constant and M_{pl} is the mass of the planet.

F_{XUV} is inversely proportional to the square of the orbital separation between the planet and its star a :

$$F_{XUV} = \frac{L_{\star}}{4\pi a^2} \quad (2)$$

where $L_{\star}$ is the XUV luminosity of the star and a is the orbital distance around the given star defined by the middle of the CHZ.

3.3 Continuously Habitable Zone

With the planets focused on in this work being in the habitable zone, we calculate their orbital distances with the CHZ. Based on the estimates from previous works that the Earth likely fully lost its primordial atmosphere by an age of ~ 100 Myr (e.g. Pepin (1991); Lammer et al. (2014)), we consider the CHZ to be the region that is habitable for the entirety of the duration of 100 Myr – 5 Gyr. Using stars ranging in mass from $0.3 M_{\odot}$ to $1.0 M_{\odot}$ in steps of 0.01, we combine the evolution of the recent Venus and early Mars limits defined by Kopparapu et al. (2013) with the luminosities and effective temperatures from the stellar evolution models developed by Spada et al. (2013) to calculate the CHZ. Shown in Figure 7, the values used for the CHZ orbital distance a in Equation 2 around each star are in the middle of the recent Venus and early Mars boundaries defining the region of continuous habitability. Their exact values around some of the considered stars are shown in Table 2.

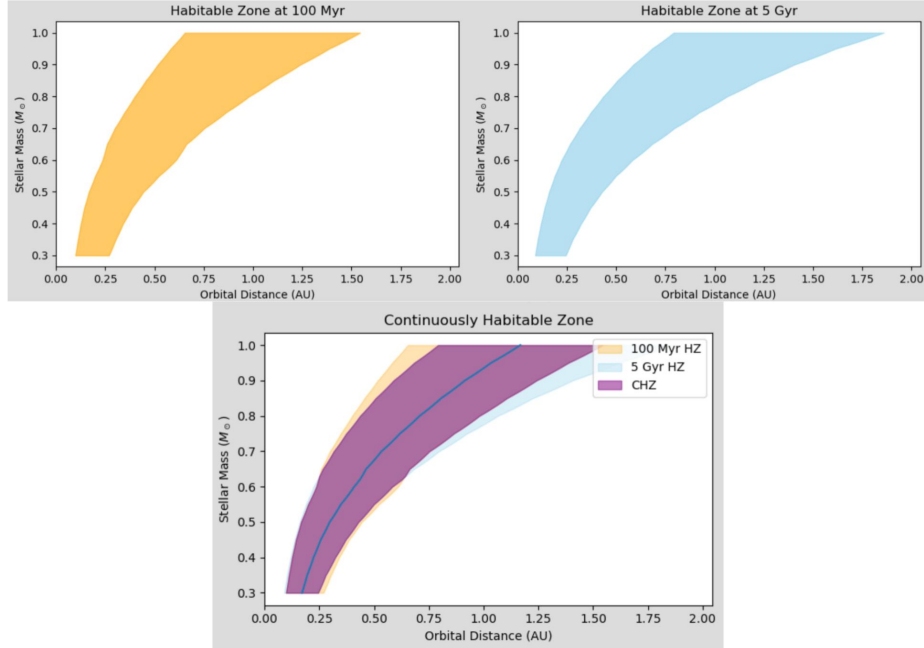


Figure 7: The overlap of the habitable zone around 71 different stars ranging in mass from $0.3 - 1.0 M_{\odot}$ at ages of 100 Myr (top left panel) and 5 Gyr (top right panel) are used to define the CHZ (bottom middle panel). The blue line in the middle of the CHZ indicates the values used for a . Their exact values for some of the stars are shown in Table 2.

²This value is consistent with models where small gas fractions ($< 5\%$) can inflate planetary radii by 10-20% (e.g. Lopez and Fortney (2014); Owen and Wu (2016)) and aligns with exoplanetary population analyses (Hadden and Lithwick (2017))

Mass ($M_{\odot}$)	a (AU)
0.3	0.171
0.4	0.224
0.5	0.298
0.6	0.410
0.7	0.534
0.8	0.708
0.9	0.915
1.0	1.168

Table 2: The values of the orbital distances to planets in the CHZ around each star. Note that only 8 stars are shown here; a total of 71 stars are considered within this range of masses.

3.4 Planetary Mass-Radius Relations

The final variables considered in the energy-limited approximation involve the characteristics of the planet at hand: M_{Pl} , R_{Pl} , and R_{XUV} . In §4.3, a range of planetary masses are considered. Their radii are calculated using the mass-radius relations from Chen and Kipping (2016) shown in the right panel of Figure 8. For consistency, the right panel of Figure 8 shows the reproduced function used in this work. The Terran Worlds’ radii go as $M^{0.28}$, the Neptunian Worlds’ go as $M^{0.59}$, and the Jovian Worlds’ go as $M^{-0.04}$. The planets used in this work fall in the Terran Worlds’ regime and are shown in Table 3.

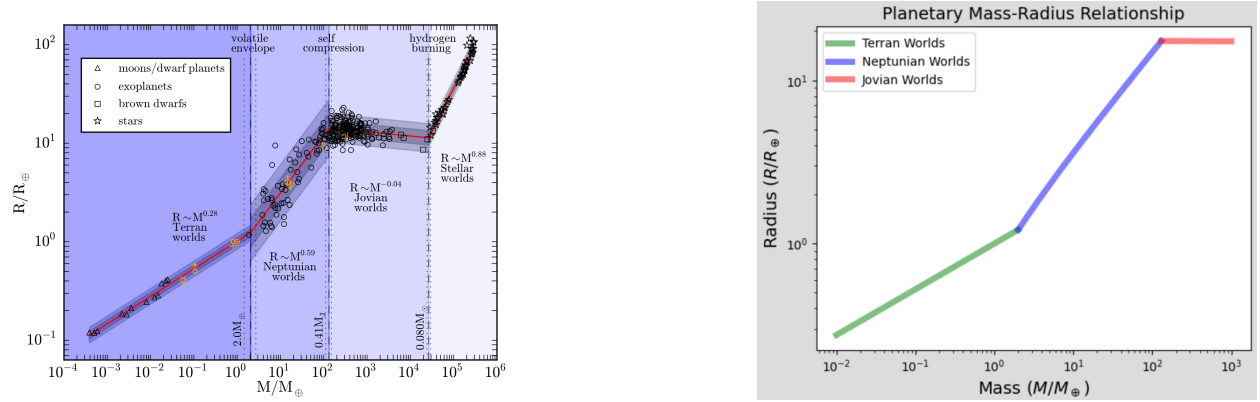


Figure 8: The right panel is a product of this work that encodes the mass-radius relation from Chen and Kipping (2016) shown in the left panel. The mass-radius relation adapted from Chen and Kipping (2016) is based on 316 objects from dwarf planets to late-type stars which are indicated in the left panel’s legend. The red line in the left panel is the mean of Chen and Kipping (2016)’s model; it is the analytical piecewise function of the mean that is reproduced in the right panel.

Mass ($M_{\oplus}$)	Radius ($R_{\oplus}$)	Mass ($M_{\oplus}$)	Radius ($R_{\oplus}$)
0.1	0.52	1.1	1.02
0.2	0.64	1.2	1.05
0.3	0.71	1.3	1.08
0.4	0.77	1.4	1.10
0.5	0.82	1.5	1.12
0.6	0.87	1.6	1.14
0.7	0.90	1.7	1.16
0.8	0.94	1.8	1.18
0.9	0.97	1.9	1.20
1.0	1.0	2.0	1.21

Table 3: The planetary masses and radii used in this work computed using the mass-radius relation adapted from Chen & Kipping (2016), rounded to two decimal places for readability.

4 Results

4.1 Mini Earth

After determining the CHZ for all of the considered stars, the energy-limited approximation of the mass-loss rate due to atmospheric escape was calculated. First, a sub-Earth (Mini Earth) similar to the one utilized by Tu et al. (2015) is placed around a solar-type star and the mass-loss rate of the planet’s atmosphere over time is modeled. Mini Earth is $0.5M_{\oplus}$, has an initial atmospheric mass of $5 \times 10^{-3}M_{\oplus}$, and orbits its $1M_{\odot}$ star at 1 AU. The radius of Mini Earth was found to be $\sim 0.82R_{\oplus}$ using the mass-radius relation for Terran Worlds found in Figure 8. The left panel of Figure 9 uses Equation 1 to compare the mass-loss rate of Mini Earth over time when orbiting stars with different initial rotational velocities. The overall shape of the plot is the same as the luminosity tracks in Figure 6 because Equation 1 is proportional to L_{XUV} . As a result, the mass-loss rate when orbiting a faster rotating star is greater than when orbiting a slower rotator. The middle panel of Figure 9 shows the atmospheric mass remaining over time for Mini Earth. The main takeaway from this figure is that a faster rotating star results in more atmospheric mass lost over time. Then, the right panel of Figure 9 shows the total mass lost in kilograms over time, further emphasizing the idea that faster stellar rotation leads to more mass lost. Although in none of these cases does Mini Earth lose all of its atmosphere, it is clear that it gets much closer to doing so with a faster rotating star.

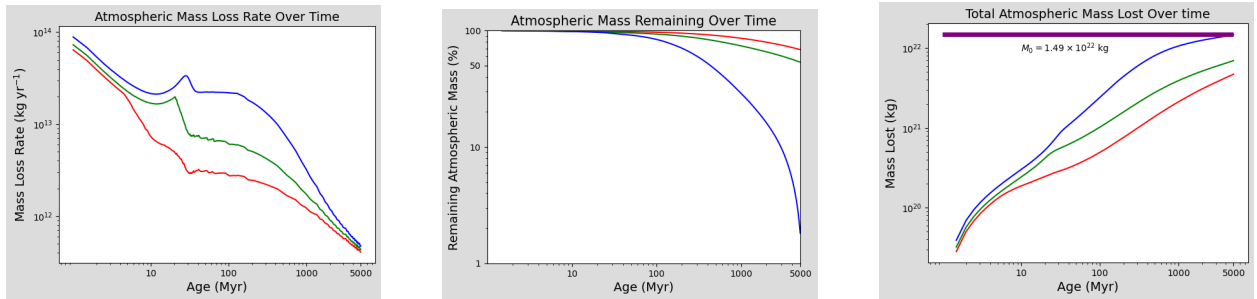


Figure 9: The evolution of the atmospheric mass-loss rate and resulting total mass lost is shown. The left panel shows the mass loss rate over time for Mini Earth around a solar mass star. The middle panel shows the fraction of atmospheric mass remaining over time. The right panel shows the total mass lost over time with the purple line indicating the initial mass of the atmosphere.

4.2 More Stars

The study was then expanded to consider four new stars with masses of 0.3, 0.5, 0.7, and $0.9 M_{\odot}$ and two intermediate rotational distribution percentiles: the 25th and 75th. An updated table, Table 4, indicates which adjective used in coming figures refers to which percentile of stellar rotation.

Adjective	Rotation Distribution Percentile
Slowest	5
Slow	25
Median	50
Fast	75
Fastest	95

Table 4: The quantitative values associated with the qualitative descriptors used throughout this report. The percentile of the model distribution from Johnstone et al. (2021) is specified in the right column with higher percentiles indicating faster initial stellar rotation.

The L_{XUV} evolutionary tracks for each star are shown in Figure 10. The fastest rotators have the highest L_{XUV} and the more massive the star the higher the L_{XUV} .

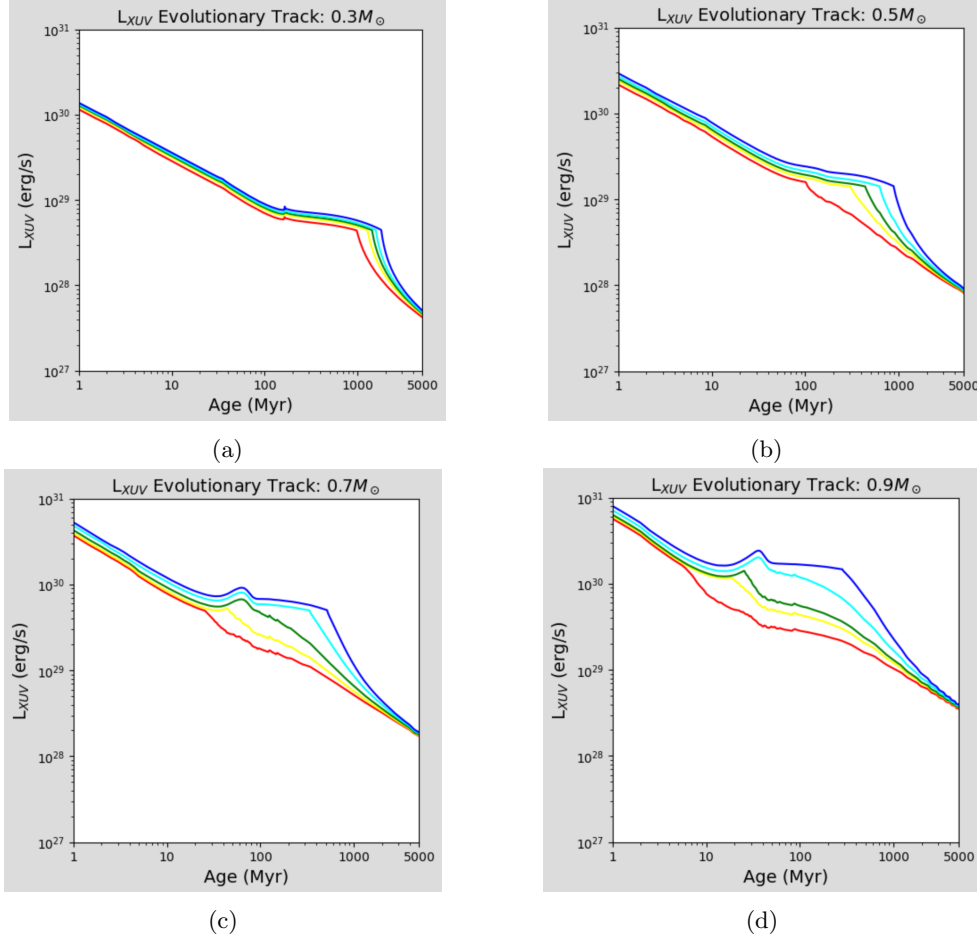


Figure 10: The L_{XUV} evolutionary tracks of stars of different masses. The stellar rotation rate indicated by the colored lines are displayed in Table 4.

Placing Mini Earth in the CHZ and calculating the atmospheric mass remaining over time reveals that despite the fact larger stars have higher luminosities, it is when a planet orbits a smaller star when atmospheric escape is at its highest. Equation 2 shows that F_{XUV} is inversely proportional to the square of the orbital separation while proportional to L_* . Therefore, F_{XUV} is more affected by a change in orbital separation than by a change in the star's luminosity. Considering Equation 1, the mass-loss rate is proportional to F_{XUV} and its relationship with orbital separation is therefore described by that of F_{XUV} 's. A comparison between Figure 11a and Figure 11d illustrates this point as Mini-Earth loses the entirety of its atmosphere before 1 Gyr regardless of stellar rotation rate in the first and retains $\gtrsim 10\%$ of its mass in all cases in the latter. The effect of stellar rotation on atmospheric escape is best seen in Figures 11b and 11c as they present a variety of outcomes that depend on the star's initial rotation rate. For example, in Figure 11b the only scenario where Mini Earth retains some of its atmosphere is with the Slowest rotator, while in Figure 11c the only stars that cause total atmospheric mass loss are the Fast and Fastest.

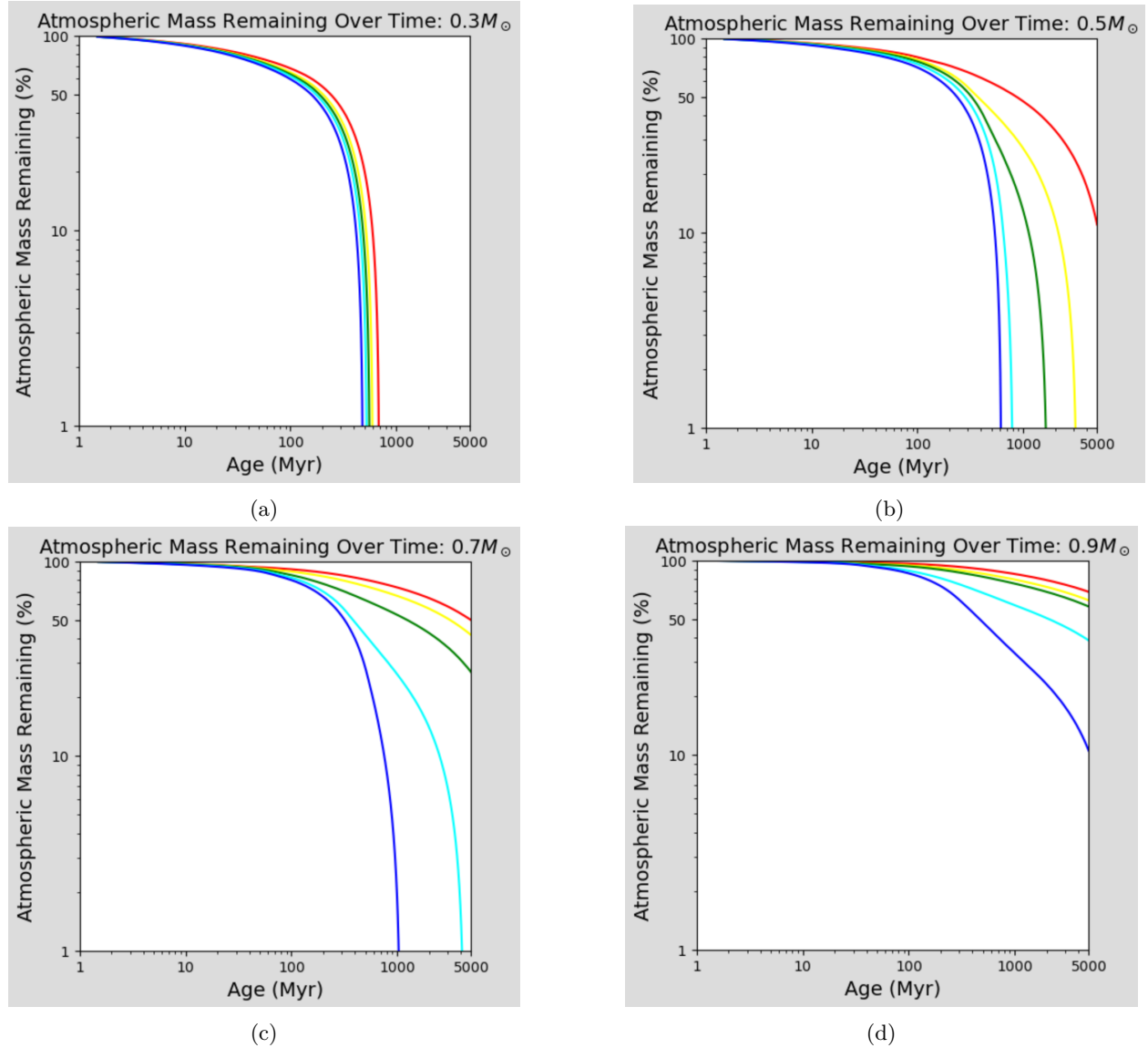


Figure 11: The percent of atmospheric mass remaining over time for Mini Earth orbiting different stars. The quantitative values for the rotational velocities of the stars associated with each can be found in Table 4.

To more closely investigate the atmospheric mass retention over time for Mini-Earth around different stellar rotators, atmospheric mass retention over time as a function of stellar mass, rotation, and age was visualized. Figure 12 shows the results for the Slowest, Median, and Fastest cases. The dark blue region of the heat maps indicates total atmospheric escape while the yellow region indicates no atmospheric mass lost. The former region gets larger as the stellar rotation increases while the latter region shrinks. The increase in verticality of the contour lines from the Slow to Fast rotators indicates that the mass of a star becomes less relevant to a planet’s ability to retain its atmosphere if the star is rotating fast enough.

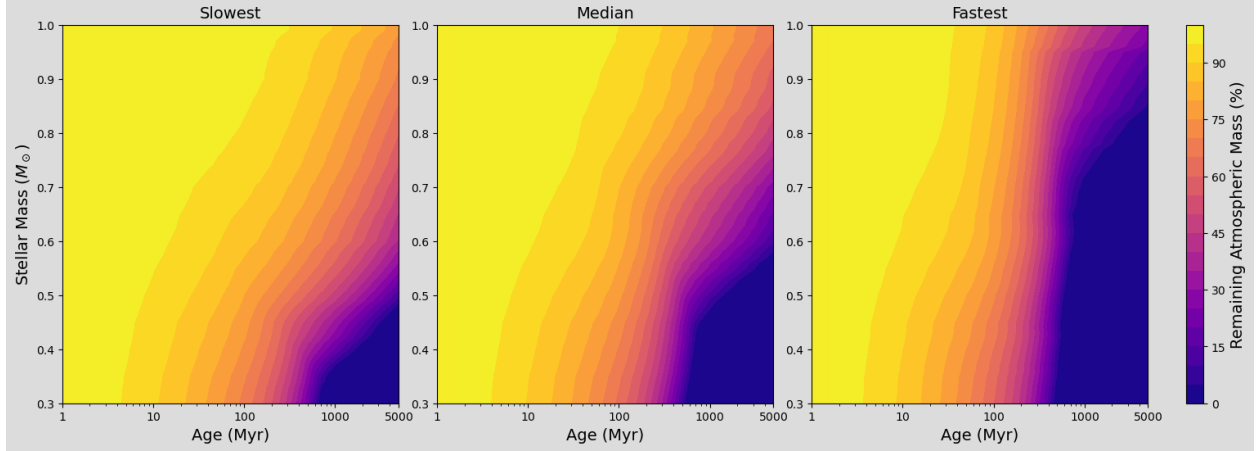


Figure 12: Atmospheric mass retention as a function of stellar age and stellar mass for terrestrial planets orbiting stars with varying initial rotation rates. Each panel shows the fraction of the planet’s initial atmosphere remaining (%) as a function of stellar mass (y-axis) and stellar age (x-axis). The three panels correspond to different stellar rotation percentiles based on the MORS rotation evolution model: Slowest rotators (5th percentile, left), Median rotators (50th percentile, middle), and Fastest rotators (95th percentile, right). The shaded yellow region indicates the earliest phase of stellar evolution, where XUV-driven escape is most intense but also when the planet has most of its initial atmosphere. Faster rotating stars lead to significantly enhanced atmospheric loss due to prolonged high XUV output during the saturation phase.

4.3 More Planets

Lastly, we consider a wider range of planets. First, we look at terrestrial planets ranging in core mass from $0.1 - 1.0M_{\oplus}$. To estimate the age at which they lose their atmospheres, the contour lines indicating only 5% atmospheric mass remaining are shown for each rotation case in Figure 13. Unreliable interpolation at levels below 5% was intentionally avoided and the difference in time between 95% and 100% mass loss is minimal.

It is evident from Figure 13 that the smaller the planet, the faster the atmospheric escape. This is likely a result of a combination of less initial atmospheric mass and lower planetary gravity. The initial atmospheric masses were 0.5% of each planet core’s mass, therefore the smaller planets had less to lose than their larger counterparts. The main takeaway from Figure 13 is that a small planet orbiting a fast-rotating, small star results in the most rapid atmospheric mass lost as calculated using the energy-limited approximation. For small stellar masses, there is little difference from Slowest to Fastest rotation rates in the age at ~total atmospheric loss, however the most noticeable differences arise in the cases of the higher-mass planets.

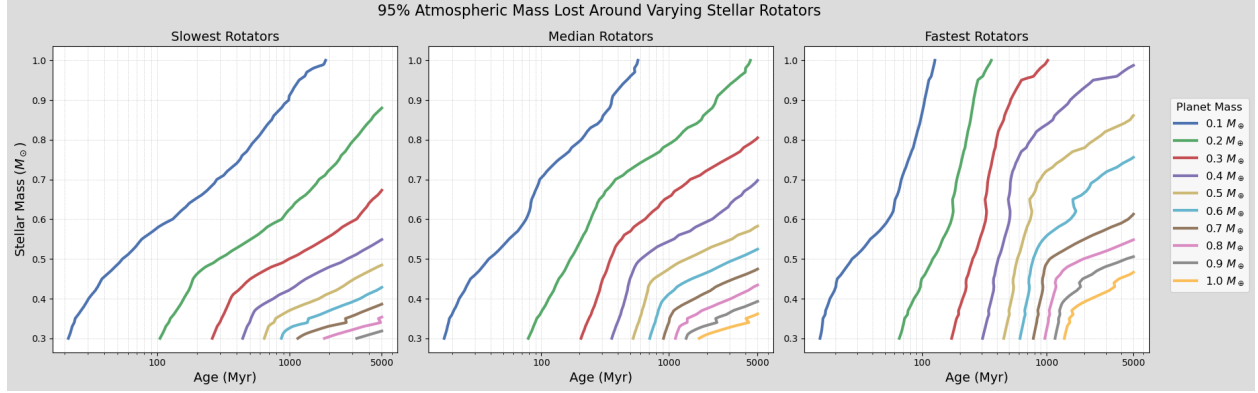


Figure 13: The ages at near-total (95%) mass loss for sub-Earth planets ranging in mass from $0.1 - 1.0 M_{\oplus}$ as a function of stellar mass and age in three cases of stellar rotation are shown. For readability only the Slowest, Median, and Fastest cases are shown. The Slow and Fast cases can be found in the Appendix. The left panel shows the Slowest (5th percentile) rotators, the middle panel the Median (50th percentile) rotators, and the right panel the Fast (95th percentile) rotators.

To place the previously explored planets into context, super-Earths ranging in core mass from $1.1 - 2.0 M_{\oplus}$ were modeled. The most telling results occurred around the Fast and Fastest rotating stars shown in Figure 14.

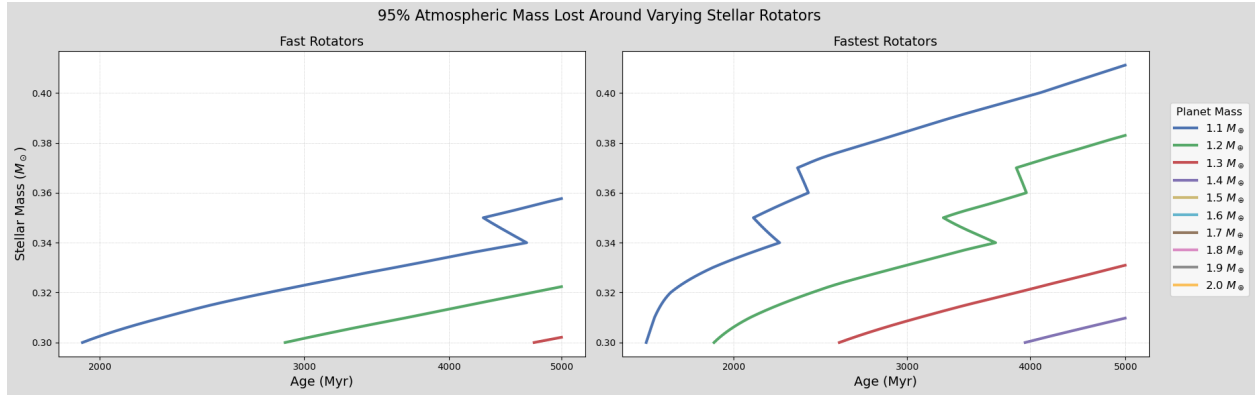


Figure 14: The ages at near-total (95%) mass loss for super-Earths ranging in mass from $1.1 - 2.0 M_{\oplus}$ as a function of stellar mass and age in two cases of stellar rotation. Only the Fast and Fastest cases are shown as none of the planets lose 95% of their atmospheric mass when orbiting any mass star rotating slower than Fast. The left panel shows the Fast (75th percentile) rotators and the right panel the Fastest (95th percentile) rotators. Note that the x- and y-axis scales are different from the previous, similar Figure 13 in order to depict more clearly the results. The ages shown span only from 1 Gyr – 5 Gyr, although they were evolved from 1 Myr to 5 Gyr just as the sub-Earths were. Also, only around low-mass stars did any super-Earths lose 95% of their atmospheric mass, hence the range of stellar masses shown on the y-axis lying between $0.3 - \sim 0.42 M_{\odot}$. Only the $1.1 - 1.4 M_{\oplus}$ planets were able to lose their atmospheres in the given time-frame. The rest of the super-Earths are still indicated in the legend to emphasize that they do not lose 95% of their atmospheric masses within the given evolution time-frame.

5 Discussion

5.1 Summary of Key Results

It was found that an exoplanet’s ability to retain its atmosphere depends on stellar mass and initial rotation rate. Specifically, it was found in Figure 7 that lower stellar mass results in a closer CHZ and subsequently higher XUV flux incident on the planet as defined in Equation 2. The resulting atmospheric loss is shown in Figure 13 and is estimated by the energy-limited approximation of hydrodynamic escape defined in Equation 1. The initial rotation rates of the stars evolved also played important roles in the evolution of the planets’ atmospheres; a comparison between the left and right panels of Figure 13 shows faster initial rotation results in more transient primordial H/He atmospheres. It was also found in Figure 10 that lower mass stars’ XUV evolutionary tracks are less sensitive to their initial rotation than their higher mass counterparts.

Both sub-Earths ($0.1 - 1.0 M_{\oplus}$) and super-Earths ($1.1 - 2.0 M_{\oplus}$) were considered. Nearly all of the sub-Earths’ atmospheres evaporated completely in all stellar mass-rotation cases, but were strongly dependent on their orbital distance and thus stellar mass. The sub-Earths on the upper end of the given range of masses were more sensitive to changes in stellar rotation than their lower mass counterparts, meaning the difference in the age of 95% atmospheric mass loss for an Earth-mass planet around a $0.3 M_{\odot}$ star varies from cases of Slow to Fast rotators more than that of a small, $0.1 M_{\oplus}$ sub-Earth as shown in Figure 13. The super-Earths were found to not lose their atmospheres in the vast majority of the stellar parameter space. The only instances of near-total evaporation for super-Earths are shown in Figure 14.

5.2 Sensitivity to Assumptions

The only driver of atmospheric escape considered in this work is photoevaporation. Jeans escape is not taken into account because it is a slow escape mechanism that dominates in cool, stable atmospheres and low XUV flux environments (e.g. Tian et al. (2008); Moore and Horwitz (2007)). Impact erosion is not considered because it has been shown that it is stochastic and less efficient overall than hydrodynamic escape, even during heavy bombardment (e.g. Genda and Abe (2003), Kegerreis et al. (2020)). Sputtering is not considered because it has been shown that sputtering is not efficient until thermal escape becomes ineffective (e.g. Johnson (1994); Leblanc and Johnson (2001)). Polar wind’s effects on atmospheric mass loss are omitted because they are relevant for heavier ions which thermal escape mechanism cannot eject easily, and only if the planet has a strong magnetic field and mature ionosphere which is unlikely during early H/He loss (e.g. Collinson et al. (2015); Collinson et al. (2016); Collinson et al. (2024)).

The escape rate in the energy-limited framework is directly proportional to the heating efficiency ϵ , which quantifies the fraction of incident high-energy XUV radiation that goes into lifting atmospheric gas out of the planet’s gravitational pull. The value $\epsilon = 0.3$ is adopted, consistent with assumptions in literature for hydrogen-dominated gaseous envelopes under high XUV flux (e.g. Lopez and Fortney (2012); Owen and Jackson (2012); Salz et al. (2016)). However, this parameter is poorly constrained and can vary depending on composition, altitude of energy deposition R_{XUV} , atmospheric density and structure, and time-dependent stellar spectra. While $\epsilon = 0.3$ offers a reasonable baseline, varying it by ± 0.2 can significantly impact an atmosphere’s ability to persist, with higher values making escape more aggressive and lower values elongating the retention time.

The initial hydrogen-helium envelope mass fraction defines the size of the primordial atmosphere at the start of the model and was set to 0.5% of the core mass of each planet. Envelope fractions can range from 0.01% to several percent for terrestrial cores making the value chosen consistent with predictions for low-mass cores (e.g. Lopez and Fortney (2012); Owen and Wu (2017)). Slight increases in envelope mass can allow a planet to retain a residual H/He layer even around active stars, while smaller envelopes are stripped in nearly all cases with sensitivity strongest near the fast-rotator, low-mass stellar boundary.

The rotational cases considered in this work (5th, 25th, 50th, 75th, and 95th percentiles from MORS; Johnstone et al. (2021)) significantly influence the duration and magnitude of the planets’ exposure to XUV radiation. The Fastest rotators exhibit the longest saturation phase and highest L_{XUV} , leading to greater envelope loss. However, the mapping from initial rotation to long-term XUV output depends on stellar age calibrations and empirical fits, which introduce some uncertainty. In particular, recent observations (e.g. Tu et al. (2015); van Saders et al. (2019)) show scatter in rotation-age-XUV relations, especially in

the pre-main-sequence phase as individual stars may deviate due to disk-locking duration, initial angular momentum, and/or magnetic activity.

The assumption made in this work of fixed core radii and envelope geometry is reasonable for low-mass planets with thin H/He envelopes, but neglects several factors including thermal contraction of the envelope and radius inflation due to XUV heating (e.g. Owen and Wu (2017)). More detailed models (e.g. Lopez and Fortney (2014); Linder et al. (2019)) predict that planetary radii shrink significantly over time, particularly in the first ~ 100 Myr. The static radius assumption made in this work therefore slightly overestimates escape at late times, since a shrinking planet presents a smaller cross-section, introducing modest conservative bias into atmospheric retention outcomes.

All calculations in this work assume that the planets reside within the CHZ around their host stars. The CHZ is defined using the evolution of the recent Venus and early Mars limits defined by Kopparapu et al. (2013) over a period of a stellar age of 100 Myr to 5 Gyr. The exact orbital distances used are in the middle of the boundaries defining the overlap of the HZ at 100 Myr and at 5 Gyr. We use the age of 100 Myr because previous works reason that Earth lost its primordial atmosphere at an age of ~ 100 Myr. The age of 5 Gyr was used based on the rationale that the planet needs to be in a habitable region for a long time after losing its primordial atmosphere for life as we know it to develop. What this assumption does not take into account is planetary migration; many planets are expected to undergo Type I migration during the gas disk phase (e.g. Baruteau and Masset (2008)) which can move a planet across XUV flux regimes or in/out of the HZ altogether during the first few Myr. Future work may focus on the planet ending up in the classically defined habitable zone at the time of total atmospheric loss rather than staying in it for its entire lifetime, allowing for planetary migration and potentially significant changes in their radiation environment.

5.3 Comparison with Previous Work

The results of this work align with and extend several key findings in the literature regarding the evolution and loss of primordial atmospheres around terrestrial exoplanets. We build upon the established energy-limited photoevaporation framework and incorporate rotation-dependent XUV luminosity histories, allowing for a more detailed examination of how stellar activity variability affects atmospheric loss.

Tu et al. (2015) demonstrated that stellar rotation history critically determines the extent of photoevaporative mass loss. Using a $1 M_{\odot}$ star and three rotational tracks (10th, 50th, and 90th percentiles of the rotational distribution from Johnstone et al. (2015)), they showed that planets around fast rotators experience much greater atmospheric loss. The results in this work reproduce this qualitative trend and extend it across a broader stellar mass range ($0.3 - 1.0 M_{\odot}$), revealing that rotation-dependent atmospheric loss becomes more extreme for planets in the HZ around low-mass stars. Unlike Tu et al., the rotation tracks implemented in this work are from MORS (Johnstone et al. (2021)) which provide finer resolution of rotational divergence at early ages and produces a more realistic stellar activity decay across mass due to coupling spin-down models with XUV luminosity evolution over time.

Luger and Barnes (2015) focused on planets in the habitable zone of M dwarfs and demonstrated that total loss of primordial hydrogen envelopes can occur due to extended pre-main-sequence high-luminosity phases and intense XUV irradiation. Their work emphasized that early runaway hydrodynamic escape could lead to abiotic O_2 accumulation. We confirm in this work that planets around M-dwarfs are the most vulnerable to hydrodynamic escape, but it is also shown that atmospheric retention depends not only on stellar mass, but also on initial rotation - a dimension not specifically explored by Luger & Barnes. Our results reveal that even with the same stellar mass, different rotation tracks lead to divergent escape histories. The presence of a retained atmosphere therefore cannot be inferred from stellar mass alone, but also requires constraints on the host star’s rotation evolution.

Owen & Wu (2013, 2017) modeled atmospheric loss using 1D hydrodynamic simulations and confirmed that energy-limited escape approximations are valid when recombination-limited or photon-limited regimes are not dominant. This work’s adoption of $\epsilon = 0.3$ lies within their empirically justified range. Our results are consistent with their prediction that planets with radii $\lesssim 1.6 R_{\oplus}$ lose most or all of their atmospheres unless their cores are large or have modest insolation.

Kubyshkina et al. (2018) introduced the Hydro-Based Approximation (HBA) which was an improvement over the energy-limited approach. The HBA interpolates between grid-based hydrodynamic escape solutions. While more physically realistic, it is difficult to apply across large parameter grids. The energy-limited

model in this work, while simplified, captures the first-order dependence of atmospheric loss on XUV flux and gravitational binding energy. However, it is likely that we overestimated escape for evolved, high-gravity planets at late times due to neglecting recombination and radiative cooling effects.

5.4 Secondary Atmospheres

The secondary atmosphere forms after the dissipation of the primordial and determines long-term climate and surface habitability (e.g. Lammer et al. (2008), Wordsworth and Kreidberg (2022)). Several mechanisms can play crucial roles in its evolution. First, giant impacts can form magma oceans that eventually crystallize, outgassing volatiles including CO_2 , H_2O , and N_2 into the atmosphere (e.g. Tonks and Melosh (1993); Elkins-Tanton (2008)). Volcanic outgassing can continue over Gyr timescales and is a dominant process on Earth and Venus, however it depends on mantle composition, volatile solubility, and the extent of the magma ocean (Schaefer and Fegley (2010)). Impact delivery can also result in degassing of volatiles from the impactor. If large enough, giant impacts can potentially generate transient steam or silicate atmospheres as well (Itcovitz et al. (2022)). Interior-exterior cycling through tectonics also plays an important role in the evolution of secondary atmospheres; plate tectonics recycle CO_2 , regulating the greenhouse effect and supporting the long-term carbon-silicate cycle (e.g. Kasting and Catling (2003); Foley and Driscoll (2016); Korenaga (2013)). They can also be lost by similar mechanisms as primordial atmospheres: hydrodynamic escape driven by stellar irradiation from young stars, impact erosion from small and large impactors, and jeans escape due to low gravitational binding energy (e.g. Lammer et al. (2020), Zahnle and Catling (2017)).

A planet’s ability to retain a secondary atmosphere is a prerequisite for long-term habitability and the requisite liquid water (e.g. Van Looveren et al. (2025); Kite and Barnett (2020)). Despite likely similar initial conditions such as comparable volatile inventories and magma oceans (e.g. Elkins-Tanton (2008); Zahnle et al. (2010); Lebrun et al. (2013)), Venus’s evolution diverged from Earth’s likely because of a runaway greenhouse effect, causing ocean loss and preventing further oceans from forming (e.g. Kasting (1988); Gillmann et al. (2022)). Earth’s other neighbor, Mars, lost its magnetic field early in its evolution leaving it more vulnerable to XUV radiation from the Sun and sputtering and ion escape caused by solar wind (Acuña et al. (1999)).

5.5 Future Work

Further study on the evaporation of primordial atmospheres around habitable zone exoplanets might include a different interpretation of the habitable zone and the orbital distances adopted for model planets. This work’s CHZ is defined such that the planet’s are in the habitable zone for nearly their entire lives. Other works may choose to set different bounds on the ages spanned by the CHZ to account for different predictions regarding the time needed for life to form. Furthermore on the topic of orbital separation, planetary migration should be taken into account. A combination of a differently evolving habitable zone and planetary migration could result in a more accurate estimate of the evolution of primordial atmospheres. Also, accounting for radius contraction and internal hydrodynamics would provide a more accurate estimate with the energy-limited approximation because it depends on the radius to which the atmosphere extends. Including internal hydrodynamics may result in more realistic results as well as a planet’s core luminosity may contribute to atmospheric escape.

6 Conclusion

The goal of this paper was to investigate atmospheric escape of exoplanets in the habitable zone under different conditions. The conditions controlled were the stellar mass and rotation rate, the planet mass, radius, and orbital separation. The orbital separation was determined by the Continuously Habitable Zone defined by the evolution of the recent Venus and early Mars limits from stellar ages of 100 Myr to 5 Gyr. The time at total loss of the primordial atmosphere was used as a scale for measuring the effects of different conditions.

The energy-limited approximation for atmospheric escape was used to estimate the mass loss rate of planets’ atmospheres. The effect of stellar mass and rotation rate on XUV luminosity and thus the atmospheric mass loss rate was explored by using MORS and its stellar and rotational evolution models (Johnstone et al.

(2021)). An exploration of the planetary parameter space was conducted by using mass-radius relations to vary the planet inputs to the energy-limited approximation. Comparisons between the resulting ages at complete atmospheric loss revealed clear dependencies on the tested parameters.

It was found that lower stellar mass and faster stellar rotation resulted in a shorter time scale for total atmospheric loss as shown in Figure 12. In Figure 13 it was also shown that lower mass planets retained their atmospheres for less time than their more massive counterparts, regardless of the stellar properties. The most effective combinations of the parameter space at evaporating planetary atmospheres is concluded to be a low mass planet orbiting a low mass, fast rotating star.

Future work could take into account planetary radius contraction, internal hydrodynamics, and planetary migration to consider a more accurate physical interpretation. Also, a more nuanced definition of the habitable zone may affect the orbital separation of the planet and the star, affecting the XUV flux incident on the planet.

References

- Acuña, M. H., Connerney, J. E., Ness, N. F., Lin, R. P., Mitchell, D., Carlson, C. W., McFadden, J., Anderson, K. A., Rème, H., Mazelle, C., Vignes, D., Wasilewski, P., and Cloutier, P. (1999). Global distribution of crustal magnetization discovered by the mars global surveyor mag/er experiment. *Science*, 284(5415):790–793.
- Aller, L. H. (1960). Primordial Chemical Composition of the Solar System. , 65:49.
- Baruteau, C. and Masset, F. (2008). Type I Planetary Migration in a Self-Gravitating Disk. , 678(1):483–497.
- Chen, J. and Kipping, D. M. (2016). Probabilistic forecasting of the masses and radii of other worlds. *The Astrophysical Journal*, 834(1):17.
- Collinson, G. A., Ergun, R. E., Andersson, L., Goldstein, D. B., Yelle, R. V., Halekas, J. S., Brain, D. A., Mahaffy, P. R., and Jakosky, B. M. (2015). Limits on the martian polar wind electric potential from maven observations. *Geophysical Research Letters*, 42(22):9105–9112.
- Collinson, G. A., Escoubet, C. P., Broiles, T. W., Strangeway, R. J., and Delcourt, D. C. (2024). Earth’s ambipolar electrostatic field and its role in ion escape. *Earth and Planetary Science Letters*, 569:117039.
- Collinson, G. A., Winningham, J. D., DiBraccio, G. A., Johnson, J. R., Mazelle, C., Leblanc, F., Modolo, R., Hara, T., Dubinin, E., and Jakosky, B. M. (2016). Strong electric fields drive ion escape from venus’ ionosphere. *Geophysical Research Letters*, 43(12):5771–5778.
- Elkins-Tanton, L. (2008). Linked magma ocean solidification and atmospheric growth for earth and mars. *Earth and Planetary Science Letters*, 271(1):181–191.
- Erkaev, N. V., Kulikov, Y. N., Lammer, H., Selsis, F., Langmayr, D., Jaritz, G. F., and Biernat, H. K. (2007). Roche lobe effects on the atmospheric loss from “Hot Jupiters”. , 472(1):329–334.
- Foley, B. J. and Driscoll, P. E. (2016). Whole planet coupling between climate, mantle, and core: Implications for rocky planet evolution. *Geochemistry, Geophysics, Geosystems*, 17(5):1885–1914.
- Fossati, L., Erkaev, N. V., Lammer, H., Cubillos, P. E., Odert, P., Juvan, I., Kislyakova, K. G., Lendl, M., Kubyshkina, D., and Bauer, S. (2017). Aeronomical constraints to the minimum mass and maximum radius of hot low-mass planets. *Astronomy & Astrophysics*, 598:A90.
- Genda, H. and Abe, Y. (2003). Survival of a proto-atmosphere through the stage of giant impacts: the mechanical aspects. *Icarus*, 164(1):149–162.
- Gillmann, C., Way, M. J., Avice, G., Breuer, D., Golabek, G. J., Honing, D., Krissansen-Totton, J., and Lammer, H. (2022). The long-term evolution of the atmosphere of venus: Processes and feedback mechanisms. *Space Science Reviews*, 218:103.
- Ginzburg, S., Schlichting, H. E., and Sari, R. (2018). Core-powered mass-loss and the radius distribution of small exoplanets. *Monthly Notices of the Royal Astronomical Society*, 476(1):759–765.
- Güdel, M. (2007). The sun in time: Activity and environment. *Living Reviews in Solar Physics*, 4(1):3.
- Hadden, S. and Lithwick, Y. (2017). Kepler Planet Masses and Eccentricities from TTV Analysis. , 154(1):5.
- Itcovitz, J. P., Rae, A. S. P., Citron, R. I., Stewart, S. T., Sinclair, C. A., Rimmer, P. B., and Shorttle, O. (2022). Reduced atmospheres of post-impact worlds: The early earth. *The Planetary Science Journal*, 3(5):115.
- Jeans, J. H. (1916). *The Dynamical Theory of Gases*. Cambridge University Press, Cambridge, UK, 4th edition.
- Johnson, R. E. (1994). Plasma-ion sputtering of an atmosphere. *Space Science Reviews*, 69(1-2):215–253.

- Johnstone, C. P., Bartel, M., and Güdel, M. (2021). The active lives of stars: A complete description of the rotation and XUV evolution of F, G, K, and M dwarfs. , 649:A96.
- Johnstone, C. P., Güdel, M., Brott, I., and Lüftinger, T. (2015). Stellar winds on the main-sequence. II. The evolution of rotation and winds. , 577:A28.
- Johnstone, C. P., Güdel, M., Stökl, A., Lammer, H., Tu, L., Rank-Lüftinger, T., and Dorfi, E. (2015). The evolution of stellar rotation and the hydrogen atmospheres of habitable-zone terrestrial planets. *The Astrophysical Journal Letters*, 815(1):L12.
- Kasting, J. F. (1988). Runaway and moist greenhouse atmospheres and the evolution of earth and venus. *Icarus*, 74(3):472–494.
- Kasting, J. F. and Catling, D. (2003). Evolution of a habitable planet. *Annual Review of Astronomy and Astrophysics*, 41:429–463.
- Kasting, J. F., Whitmire, D. P., and Reynolds, R. T. (1993). Habitable zones around main sequence stars. *Icarus*, 101:108–128.
- Kegerreis, J. A., Eke, V. R., Massey, R. J., and Teodoro, L. F. A. (2020). Atmospheric erosion by giant impacts onto terrestrial planets. *The Astrophysical Journal*, 897(2):161.
- Kislyakova, K. G., Johnstone, C. P., Scherf, M., Holmstrom, M., Alexeev, I. I., Lammer, H., Khodachenko, M. L., and Güdel, M. (2020). Evolution of the earth’s polar outflow from mid-archean to present. *Journal of Geophysical Research: Space Physics*, 125(9):e2020JA027837.
- Kite, E. S. and Barnett, M. N. (2020). Exoplanet secondary atmosphere loss and revival. *Proceedings of the National Academy of Science*, 117:18264–18271.
- Koenigl, A. (1991). Disk Accretion onto Magnetic T Tauri Stars. , 370:L39.
- Kopparapu, R. K., Ramirez, R., Kasting, J. F., Eymet, V., Robinson, T. D., Mahadevan, S., Terrien, R. C., Domagal-Goldman, S., Meadows, V., and Deshpande, R. (2013). Habitable Zones around Main-sequence Stars: New Estimates. , 765(2):131.
- Korenaga, J. (2013). Initiation and evolution of plate tectonics on earth: Theories and observations. *Annual Review of Earth and Planetary Sciences*, 41(Volume 41, 2013):117–151.
- Krenn, A. F., Fossati, L., Kubyshkina, D., and Lammer, H. (2021). A critical assessment of the applicability of the energy-limited approximation for estimating exoplanetary mass-loss rates. , 650:A94.
- Kubyshkina, D., Fossati, L., Erkaev, N. V., Cubillos, P. E., Johnstone, C. P., Kislyakova, K. G., Lammer, H., Lendl, M., and Odert, P. (2018). Overcoming the limitations of the energy-limited approximation for planet atmospheric escape. *The Astrophysical Journal Letters*, 866(2):L18.
- Lammer, H., Chassefière, E., Karatekin, O., Morschhauser, A., Niles, P. B., Mousis, O., Odert, P., Möstl, U. V., and Breuer, D. (2015). Outgassing history and escape of the martian atmosphere and water inventory. *Planetary and Space Science*, 109-110:139–148.
- Lammer, H., Kasting, J., Chassefière, E., Johnson, R., Kulikov, Y., and Tian, F. (2008). Atmospheric escape and evolution of terrestrial planets and satellites. *Space Science Reviews*, 139(1-4):399–436. Funding Information: Acknowledgements Helmut Lammer, James F. Kasting, Eric Chassefière and Yuri N. Kulikov thank the Helmholtz-Gemeinschaft as this research has been supported by the Helmholtz Association through the research alliance “Planetary Evolution and Life”. Yu. Kulikov and H. Lammer acknowledge also support by the Austrian Academy of Sciences, “Verwaltungsstelle für Auslandsbeziehungen”, by the Russian Academy of Sciences (RAS), for supporting working visits to the PGI/RAS in Murmansk, Russian Federation. H. Lammer and Yu. N. Kulikov also acknowledge the International Space Science Institute (ISSI; Bern, Switzerland) and the ISSI team “Evolution of Exoplanet Atmospheres and their Characterization”. R.E Johnson acknowledges the support of NASA’s Planetary Atmospheres Program.

- Lammer, H., Scherf, M., Kurokawa, H., Ueno, Y., Burger, C., Maindl, T., Johnstone, C. P., Leitzinger, M., Benedikt, M., Fossati, L., Kislyakova, K. G., Marty, B., Avicé, G., Fegley, B., and Odert, P. (2020). Loss and Fractionation of Noble Gas Isotopes and Moderately Volatile Elements from Planetary Embryos and Early Venus, Earth and Mars. , 216(4):74.
- Lammer, H., Stökl, A., Erkaev, N. V., Dorfi, E. A., Odert, P., Güdel, M., Kulikov, Y. N., Kislyakova, K. G., and Leitzinger, M. (2014). Origin and loss of nebula-captured hydrogen envelopes from ‘sub’- to ‘super-earths’ in the habitable zone of sun-like stars. *Monthly Notices of the Royal Astronomical Society*, 439(4):3225–3238.
- Lammer, H., Zerkle, A. L., Gebauer, S., Tosi, N., Noack, L., Scherf, M., Pilat-Lohinger, E., Güdel, M., Grenfell, J. L., Godolt, M., and Nikolaou, A. (2018). Origin and evolution of the atmospheres of early Venus, Earth and Mars. , 26(1):2.
- Leblanc, F. and Johnson, R. E. (2001). Sputtering of the martian atmosphere by solar wind pick-up ions. *Planetary and Space Science*, 49(6):645–656.
- Lebrun, T., Massol, H., Chassefière, E., Davaille, A., Marcq, E., Sarda, P., Leblanc, F., and Brandeis, G. (2013). Thermal evolution of an early magma ocean in interaction with the atmosphere. *Journal of Geophysical Research: Planets*, 118(6):1155–1176.
- Linder, E. F., Mordasini, C., Mollière, P., Marleau, G.-D., Malik, M., Quanz, S. P., and Meyer, M. R. (2019). Evolutionary models of cold and low-mass planets: Cooling curves, magnitudes, and detectability. *Astronomy & Astrophysics*, 623:A85.
- Lopez, E. D. and Fortney, J. J. (2012). The role of core mass in controlling evaporation. *ApJ*, 761:59.
- Lopez, E. D. and Fortney, J. J. (2014). Understanding the mass–radius relation for sub-neptunes: Radius as a proxy for composition. *The Astrophysical Journal*, 792(1):1.
- Luger, R. and Barnes, R. (2015). Extreme water loss and abiotic O_2 buildup on planets throughout the habitable zones of m dwarfs. *Astrobiology*, 15(2):119–143.
- Matt, S. P., Pinzón, G., de la Reza, R., and Greene, T. P. (2010). Spin evolution of accreting young stars. i. effect of magnetic star–disk coupling. *The Astrophysical Journal*, 714(2):989.
- Meadows, V. S., Reinhard, C. T., Arney, G. N., Parenteau, M. N., Schwieterman, E. W., Domagal-Goldman, S. D., Lincowski, A. P., Stapelfeldt, K. R., Rauer, H., DasSarma, S., Hegde, S., Narita, N., Deitrick, R., Lustig-Yaeger, J., Lyons, T. W., Siegler, N., and Grenfell, J. L. (2018). Exoplanet biosignatures: Understanding oxygen as a biosignature in the context of its environment. *Astrobiology*, 18(6):630–662.
- Melosh, H. J. and Vickery, A. M. (1989). Impact erosion of the primordial atmosphere of mars. *Nature*, 338(6215):487–489.
- Moore, T. E. and Horwitz, J. L. (2007). Stellar ablation of planetary atmospheres. *Reviews of Geophysics*, 45(3):RG3002.
- Odert, P., Erkaev, N. V., Kislyakova, K. G., Lammer, H., Mezentsev, A. V., Ivanov, V. A., Fossati, L., Leitzinger, M., Kubyshkina, D., and Holmström, M. (2020). Modeling the ly transit absorption of the hot jupiter hd 189733b. *AA*, 638:A49.
- Owen, J. E. (2019). Atmospheric escape and the evolution of close-in exoplanets. *Annual Review of Earth and Planetary Sciences*, 47:67–90.
- Owen, J. E. and Alvarez, M. A. (2015). Uv driven evaporation of close-in planets: Energy-limited, recombination-limited, and photon-limited flows. *The Astrophysical Journal*, 816(1):34.
- Owen, J. E. and Jackson, A. P. (2012). Planetary evaporation by uv and x-ray radiation: basic hydrodynamics: Planetary evaporation. *Monthly Notices of the Royal Astronomical Society*, 425(4):2931–2947.

- Owen, J. E. and Wu, Y. (2013). Kepler planets: A tale of evaporation. *The Astrophysical Journal*, 775(2):105.
- Owen, J. E. and Wu, Y. (2016). Atmospheres of low-mass planets: The “boil-off”. *The Astrophysical Journal*, 817(2):107.
- Owen, J. E. and Wu, Y. (2017). The evaporation valley in the kepler planets. *The Astrophysical Journal*, 847(1):29.
- Parker, E. N. (1958). Dynamics of the interplanetary gas and magnetic fields. *Astrophysical Journal*, 128:664–676.
- Pepin, R. O. (1991). On the origin and early evolution of terrestrial planet atmospheres and meteoritic volatiles. *Icarus*, 92(1):2–79.
- Pizzolato, N., Maggio, A., Micela, G., Sciortino, S., and Ventura, P. (2003). The stellar activity-rotation relationship revisited: Dependence of saturated and non-saturated X-ray emission regimes on stellar mass for late-type dwarfs. , 397:147–157.
- Prinn, R. G. (1982). Origin and evolution of planetary atmospheres: An introduction to the problem. , 30(8):741–753.
- Ramirez, R. M. (2018). A more comprehensive habitable zone for finding life on other planets. *Geosciences*, 8(8).
- Ribas, I., Guinan, E. F., Güdel, M., and Audard, M. (2005). Evolution of the Solar Activity over Time and Effects on Planetary Atmospheres. I. High-Energy Irradiances (1-1700 Å). , 622(1):680–694.
- Salz, M., Czesla, S., Schneider, P. C., and Schmitt, J. H. M. M. (2016). Simulating the escaping atmospheres of hot gas planets in the solar neighborhood. , 586:A75.
- Schaefer, L. and Fegley, B. (2010). Chemistry of atmospheres formed during accretion of the earth and other terrestrial planets. *Icarus*, 208(1):438–448.
- Schunk, R. (1986). An updated theory of the polar wind. *Advances in Space Research*, 6(3):79–88.
- Solomon, S. C. and Head, J. W. (1991). Fundamental issues in the geology and geophysics of venus. *Science*, 252(5003):252–260.
- Spada, F., Demarque, P., Kim, Y. C., and Sills, A. (2013). The Radius Discrepancy in Low-mass Stars: Single versus Binaries. , 776(2):87.
- Tian, F., Kasting, J. F., Liu, H.-L., and Roble, R. G. (2008). Hydrodynamic planetary thermosphere model: 1. Response of the Earth’s thermosphere to extreme solar EUV conditions and the significance of adiabatic cooling. *Journal of Geophysical Research (Planets)*, 113(E5):E05008.
- Tian, F., Toon, O. B., Pavlov, A. A., and De Sterck, H. (2005). Transonic hydrodynamic escape of hydrogen from extrasolar planetary atmospheres. *The Astrophysical Journal*, 621:1049–1060.
- Tonks, W. B. and Melosh, H. J. (1993). Magma ocean formation due to giant impacts. *Journal of Geophysical Research: Planets*, 98(E3):5319–5333.
- Tu, L., Johnstone, C. P., Güdel, M., and Lammer, H. (2015). The extreme ultraviolet and X-ray Sun in Time: High-energy evolutionary tracks of a solar-like star. , 577:L3.
- Van Looveren, G., Boro Saikia, S., Herbort, O., Schleich, S., Güdel, M., Johnstone, C., and Kislyakova, K. (2025). Habitable Zone and Atmosphere Retention Distance (HaZARD): Stellar-evolution-dependent loss models of secondary atmospheres. , 694:A310.
- van Saders, J. L., Pinsonneault, M. H., and Barbieri, M. (2019). Forward Modeling of the Kepler Stellar Rotation Period Distribution: Interpreting Periods from Mixed and Biased Stellar Populations. , 872(2):128.

- Volkov, A. N., Tucker, O. J., Erwin, J. T., and Johnson, R. E. (2011). Thermally driven atmospheric escape: Transition from hydrodynamic to jeeps escape. *The Astrophysical Journal Letters*, 729(2):L24.
- Watson, A. J., Donahue, T. M., and Walker, J. C. G. (1981). The dynamics of a rapidly escaping atmosphere: Applications to the evolution of Earth and Venus. , 48(2):150–166.
- Wordsworth, R. and Kreidberg, L. (2022). Atmospheres of rocky exoplanets. *Annual Review of Astronomy and Astrophysics*, 60(Volume 60, 2022):159–201.
- Yau, A. W., André, M., and Peterson, W. K. (2007). Polar wind and magnetospheric plasma. *Journal of Atmospheric and Solar-Terrestrial Physics*, 69(17-18):1936–1951.
- Zahnle, K., Schaefer, L., and Fegley, B. (2010). Earth’s earliest atmospheres. *Cold Spring Harbor Perspectives in Biology*, 2(10):a004895.
- Zahnle, K. J. and Catling, D. C. (2017). The cosmic shoreline: The evidence that escape determines which planets have atmospheres, and what this may mean for proxima centauri b. *The Astrophysical Journal*, 843(2):122.